MILITARY SPECIFICATION

OPERATIONS UNIT, FOR ROCKET MANAGEMENT SUBSYSTEM, M138

This specification is approved for use by the U.S. Army Armament, Munitions and Chemical Command and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers requirements, quality assurance provisions, and preparation for delivery for Operations Unit, for Rocket Management Subsystem, M138.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. Unless otherwise specified (see 6.2), the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-E-5400	-	Electronic Equipment, Airborne, General Specification for
MIL-W-63150	-	Weapons and Support Material, Standard Quality Assurance Provisions for

STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes

FSC 1310

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Center, Attn. DRSMC-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MIL-STD-280	-	Definitions of Item Levels, Item Exchangeability, Models and Related Terms
MIL-STD-810	-	Environmental Methods (with Notice 2 (11))
MIL-STD-1235	-	Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings and publications form a part of this specification to the extent specified herein:

DRAWINGS

US ARMY ARMAMENT RESEARCH AND DEVELOPMENT CENTER (ARDC)

PRODUCT

9324105	-	OU Interface Control Drawing
9324108	-	Operations Unit
9324400	-	DU/OU Interface Specification, RMS
12011785	-	Burn-In Test Procedure

PACKAGING

P9324108	-	Operations Unit

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM-D-3951	-	Standard Practice for Commercial Practice

(Copies of specifications, standards, handbooks, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Item definition. The Operations Unit (OU) is a Line Replaceable Unit (LRU) in the M138 Rocket Management Subsystem. The OU is capable of identifying launcher types, inventorying 2.75 inch rockets, setting resistance capacitance fuzes and firing

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rockets upon command from the Display Unit (DU). Major components of the OU are: Ohmmeter/Setter Assembly, Sequencer and I/O Assembly, Power Supply and Built-In-Test Circuit Assembly, Baseplate Assembly, and Fuze Set/Squib Fire Assembly.

3.2 First article inspection. This specification contains technical provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract and 4.4.

3.3 Materials. Materials shall be in accordance with applicable drawings, specifications and standards.

3.4 Design and construction. The M138 Operations Unit shall comply with all requirements specified on Drawing (Dwg) 9324108, all associated drawings, and with all requirements specified in applicable specifications and standards. The OU shall be constructed to meet the requirements of MIL-E-5400, for Class 1A equipment.

3.4.1 Interchangeability. All Operational Units manufactured to this specification shall be functionally interchangeable as defined in MIL-STD-280. Interchangeability shall exist between units and all replaceable assemblies, subassemblies, and parts having the same part number.

3.4.2 Preconditioning (burn-in). The "burn-in" shall be conducted in accordance with, Burn-in Test Procedure, 12011785.

3.4.3 Performance characteristics.

3.4.3.1 Performance. The performance characteristics of the OU shall be as defined in the following paragraphs.

3.4.3.1.1 Electrical interface. All electrical interface with the OU shall be through two electrical connectors as specified on 9324105, OU Interface Control Drawing.

3.4.3.1.2 Display unit interface. All electrical interface with the DU shall be through OU connector J1 as defined by Table I and 9324400 DU/OU Interface Document.

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TABLE 1. Operations unit connector J1

Pin #	Signal	Input Output	Low "O"State	High "1" State
1	NRSD		See 9324400	
2	RSD		DU/OU Interface Specification	
3	SEQ RST	OUT	.4V@ I_O =.4ma	4V min @ Z.=1K
4	-24V(TP)			
5	V COMP	OUT	.4V@I_O=5.34ma	2.7V@I_{IN}=-200μa
6	V+			
7	NSTB	OUT	.4V@I_O=4.2ma	2.4V@I_{IN}=-260μa
8	XRD	OUT	.4V@I_O=3.6ma	2.5V@I_{IN}=-350μa
9	NMEMD	IN	.4V@I_{IN}=-.86ma (.7V max)	2V min or open (10K p.u.)
10	NID	OUT	.5V@I_O=45.35ma	2.9V min @ I_O = -250μa
11	BSMAP	OUT	.5V@I_O=3.6ma	2.9V min @ I_O = -20μa
12	BLWL	OUT	.5V@I_O-2.97ma	2.9V min @ I_O = -20μa
13	-24V(TP)			
14	NCHO	OUT	.4V@I_O=1.4ma	2.4V@I_O= -70μa
15	433T	OUT	.4V@I_O=1.3ma	3V min @I_O= -25μa
16	STBY ARM	IN	GND OR OPEN	28V
17	PD	OUT	.4V@I_O=1ma	2.4V@I_O=-70μa
18	PRTN		POWER RETURN	
19	SRTN		SIGNAL RETURN	
20	XESI RST	IN	.4V max@I_{IN}= O	2V@I_{IN}=300μa
21	OHM TST		(TEST LINE: 5Ω,10Ω OR 150mV OUT)	
22	SET TST		(INTERNAL TEST POINT)	
23	BIT RD	OUT	.4V@I_O=1.4 ma	2.4V@I_O= -50μa

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TABLE 1. Operations unit connector J1 (cont'd)

24	CS-1	OUT	.4V@I_O=3.5 ma	3.5V min @I_O= -350μa
25	NMR	OUT	.4V@I_O=4.2 ma	2.7V@I_O= -90μa
26	CLK	OUT	.4V@I_O=1.58ma	4.4V@I_O= -5μa 3.5Vmin)
27	N.C.			
28	STA ADDR LSB	IN	.4V@I_{IN}= -.7ma	2V min or open circuit (10K p.u)
29	STA ADDR MSB	IN	.4V@I_{IN}= -.7ma	2V min or open circuit (10K p.u)
30	FP-2	OUT	.4V@I_O=1.6ma	2.4V@I_O= -50μa
31	CS-2	OUT	.4V@I_O=3.5ma	3.5V min @ I_O = -350μa
32	N.C.			
33	SRTN			
34	BSTBY	OUT	APPROXIMATELY OV	27.4V
35	ND	OUT	.4V@I_O=.2ma	2.4V@I_O= -40μa
36	MAP	IN	(28V INPUT SWITCHED PEAK, 10mSEC)	@ I = 10A MAX
37	ESI	OUT	OPEN (FLOATING)	+28V AIRCRAFT POWER
38	MR	OUT	.4V@I_O=2.78ma	3.5V min @ I_O = -6μa
39	NVHAZ	OUT	.4V@I_O=31ma	2.9V min @I_O = -240μa
40	NGFLT	OUT	.4V@I_O=50ma	2.9V min @I_O = -260μa
41	OHMD	OUT	.4V@I_O=1.7ma	APPROXIMATELY 2V @ I_O = 0
42	PO-O	OUT		
43	PD-1	OUT	.4V@I_O=.3ma	3.5V min @ I_O = -10μa

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TABLE 1. Operations unit connector J1 (cont'd)

44	PO-2	OUT	.4V@I_O=.3ma	3.5V min @ I_O = -10μa
45	PO-3	OUT	.4V@I_O= .3ma	3.5V min @I_O = -10μa
46	PO-4	OUT	.4V@I_O= .3ma	3.5V min @I_O = -20μa
47	PO-5	OUT		
48	PO-6	OUT		
49	PO-7	OUT		
50	BMAP	OUT	SIGNAL GND	
51	BFO	OUT	.4V@I_O= 2.97ma	2.9V min @I_O = -20μa
52	439T	OUT	.4V@I_O= 1.3ma	3V min @I_O = -25μa
53	AO	OUT	.4V@I_O= 3.19ma	3.9V min @I_O = -990μa (2V min)
54	Al	OUT		
55	A2	OUT		
56	A3	OUT		
57	A4	OUT		
58	A5	OUT		
59	A6	OUT		
60	A7	OUT		
61	DEC A0	OUT	.4V@I_O= 1.44ma	2.4V min @I_O = -8 0μa (2V min)
62	DEC A1	OUT		
63	DEC A2	OUT		
64	XMIT	OUT	.4V@I_O= .2ma	2.4V min @I_O = -60μa
65	RECV	OUT	.4V@I_O= .2ma	2.4V min @I_O = -60μa

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TABLE 1. Operations unit connector J1 (cont'd)

66	RRD	OUT	.4V@I_O= 1.3ma	3.5V min @Z_O = 10K
67	SFD	OUT	.4V@I_O= 1.3ma	3.5V min @Z_O = 10K
68	D7	OUT	.4V@I_O= 1.5ma	3.5V min @I_O = -80μa
69	FP-1	OUT	.4V@I_O= 1.6ma	2.4V min @I_O = -50μa
70	DR	OUT	.4V@I_O= 40μa	2.9V min @I_O = -20μa
71	STA ADR RTN		GND	
72	S24V			24V
73	S24V DRI	OUT	1.6V max	24V
74	A8	OUT		
75	A9	OUT	.4V@I_O= 3.19ma	3.9V min @I_O = -990μa (2V min)
76	A10	OUT		
77	A11	OUT	.4V@I_O= 3.2ma	3.9V min @I_O = -1ma
78	20V(TP)	OUT		
79	-20V(TP)	OUT		
80	+5V(TP)	OUT		
81	+12V(TP)	OUT		
82	DO	OUT		
83	D1	OUT	.4V@I_O= 1.5ma	3.5V min @I_O = -80μa
84	D2	OUT		
85	D3	OUT		
86	D4	OUT		
87	D5	OUT	.4V@I_O= 1.5ma	3.5V min @I_O = -80μa
88	D6	OUT		

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3.4.3.1.1.2.1 Electrical power. Electrical power for the OU shall be provided by the DU through connector J1. Power requirements shall be as follows:

a. OU supply voltage (VOU): +20VDC min, +21.5VDC max at 500 ma steady state, 713ma max during fuze set/rocket fire cycle.

b. Standby power (STBY): 22.5VDC min, 30VDC max at 50 ma max.

c. Master arming power (MAP): 22.5VDC min, 30VDC max at 10 amp max for 10 milliseconds firing pulses.

3.4.3.1.1.3 Rocket launcher interface. All electrical interface with the rocket launcher shall be made through connector J2 (TABLE II). Signals to take inventory, set RC fuzes, ignite rocket motors, and identify launcher type, shall be supplied through this connector.

TABLE II. Operations unit connector J2

Pin #		Function	Characteristics
1	SQ1		
2	SQ3		
3	SQ5		
4	SQ7		
5	SQ9	Squib Fire Lines	Output 5A Min.,
6	SQ11		10 mSec pulse across
7	SQ13		1 OHM load.
8	SQ15		See para.
9	SQ17		3.4.3.1.1.3.1
10	FZ1		
11	FZ3		
12	FZ5		
13	FZ7		
14	FZ9	Fuze Charge Lines	See para
15	FZ11		3.4.3.1.1.3.2 &
16	FZ13		figure 2.
17	FZ15		
18	FZ17		
19	SQ2		

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TABLE II. Operations unit connector J2 (cont'd)

20	SQ4		
21	SQ6		
22	SQ8		
23	SQ10	Squib Fire Lines	Output 5A Min.,
24	SQ12		10 mSec pulse across
25	SQ14		1 OHM load. See
26	SQ16		para 3.4.3.1.1.3.1
27	SQ18		
28	FZ2		
29	FZ4		
30	FZ6		
31	FZ8		
32	FZ10	Fuze Charge Lines	See para
33	FZ12		3.4.3.1.1.3.2 &
34	FZ14		Figure 2.
35	FZ16		
36	FZ18		
37	OHM CKT RTN (Squib RTN)		Return lines for
38	OHM CKT RTN (Squib RTN)		Squib Fire Lines
39			
40			
41	NC		
42			
43			
44	SQ19	Squib Fire Lines	See para 3.4.3.1.1.3.1
45			
46			
47	NC		
48			
49			

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TABLE II. Operations unit connector J2 (cont'd)

50			
51			
52	FZ19	Fuze Charge Line	See para 3.4.3.1.1.3.2 & Figure 2
53	OHM	OHM(TP) Ohmmeter Input	150mv, 15 OHM impedance
54			
55	NC		
56			
57			
58	CS3	Channel Sel. Line (TP)	0.4v @ 8ma, 2.4v @ .380μa
59	NC		
60	NUR	Not UART Sel.(TP)	0.4v @ 3.6ma, 2.4v @ .380μa
61	TBRL	XMIT Reg load(TP)	0.4v @ 4.0ma, 2.4v @ 400μa
62	LWL	Launcher O.D. Input	1v max @ 6.3ma, 28vDC (open Cir)
63	FO	Launcher O.D. Input	1v max @ 6.3ma, 28vDC (open Cir)
64	Launcher	Shld I.D. Input	Normally GND, 10 max when tested
65	Launcher Shld GND		Tie to Aircraft GND
66	BESI SET	ESI Set & Reset	Normally 28vDC through ESI coil.
67	BESI RSET		1v max 50 mSec when active.
68	NC		
69	Chassis RTN		

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TABLE II. Operations unit connector J2 (cont'd)

70	Chassis RTN	
71	SRTN	
72	OHM RTN (TP)	Return line for Ohm-meter
73		
74		
75		
76		
77	NC	
78		
79		
80		
81		
82		
83		
84	ECHG(TP)	Fuze Charge Monitor
85	+11v(TP)	Ohmmeter PWR 11.5v to 11.8v
86	BFP(TP)	Fire pulse monitor (all channels)
87	OHM CKTR TRN (Squib RTN)	Return PT. for all Squibs*
88	Fuze CKT RTN	Return PT. for Fuze Charge Lines*.
		*MUST BE TIED TO AIRCRAFT FRAME

3.4.3.1.1.3.1 Rocket squib lines. The OU shall provide 19 rocket squib lines and a single return line for the purpose of taking inventory and firing 2.75 inch rockets.

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3.4.3.1.1.3.1.1 Squib grounding. Each squib firing line shall be electrically grounded to the squib return line within the OU, except when it is selected for firing or interrogation. The resistance of any grounded squib circuit to the squib return shall not exceed 0.1 OHM when measured at connector J2. Sustained grounding of a squib firing line shall not impair the operation of other squib lines.

3.4.3.1.1.3.1.2 Squib interrogation. The OU shall be capable, upon command from the DU, of sequentially interrogating each squib line for the presence of a rocket squib. Detection shall be accomplished by making a resistance measurement using a circuit having an internal resistance of not more than 15 OHMS and a test current of not more than 10 Milliamperes per squib line. A resistance of less than 5.0 OHM shall constitute presence of a squib. A resistance of greater than 10 OHMS shall constitute the absence of a squib.

3.4.3.1.1.3.1.3 Squib firing pulse. The OU shall be capable upon command of applying a firing pulse to any particular squib line. This signal shall be a pulse of at least 5 amperes at 10 milliseconds minimum duration when measured across a 1 OHM load.

3.4.3.1.1.3.2 Fuze charge lines. The OU shall provide 19 fuze charge lines and a single return line for the purpose of setting M433 and M439 fuzes.

3.4.3.1.1.3.2.1 Fuze grounding. Each fuze charge line shall be electrically grounded within the OU to the fuze charge return line except when the corresponding squib line is selected for firing or interrogation. The resistance of a grounded fuze line to the return shall not exceed .275 OHMS when measured at connector J2. Sustained grounding of a fuze charge line shall not interfere with the operation of any other fuze charge line.

3.4.3.1.1.3.2.2 Fuze charge signals. The OU shall apply a charge signal similar to that shown in Figure 1. The signal shall have a maximum voltage excursions of +20V and -22V and duration of not more than 50 milliseconds.

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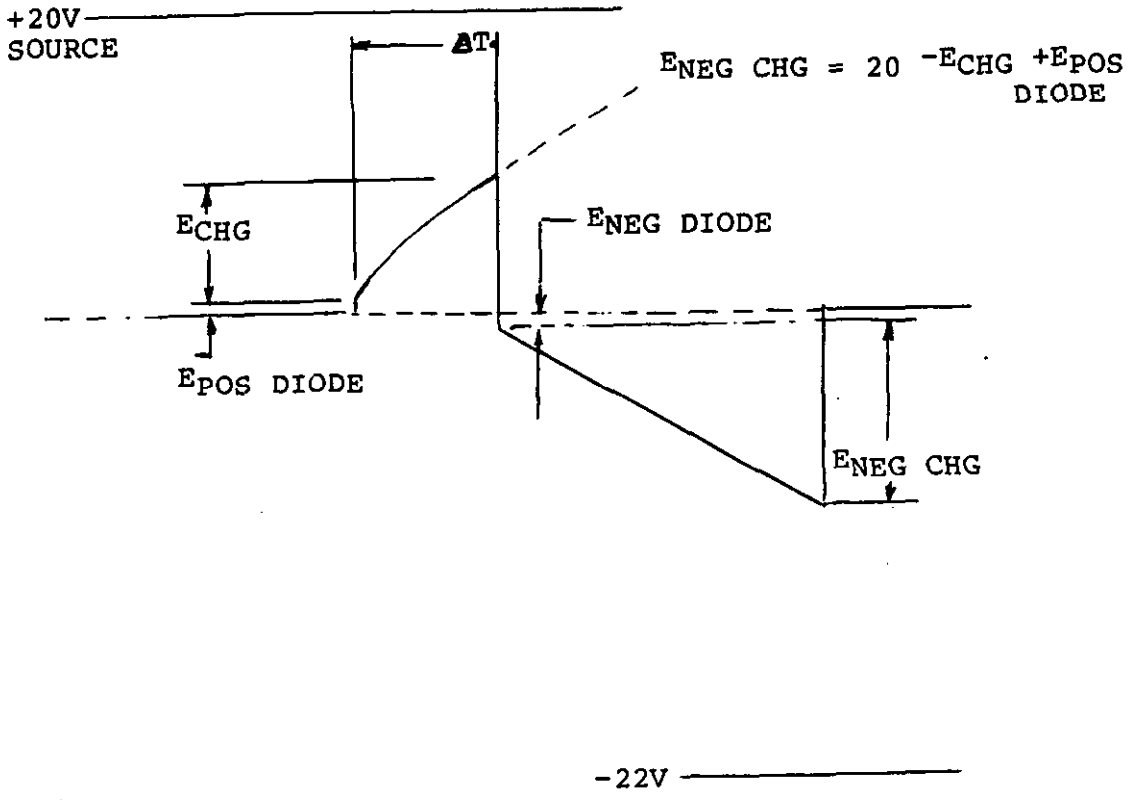


FIGURE 1. Fuze charge signal.

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3.4.3.1.1.3.2.3 The OU shall be capable of setting type M433 and M439 RC fuzes to operate within the specified accuracy limits at the time commanded by the DU (reference 9324400). The function time range for the M433 shall be 0 to 600 milliseconds. The range of function times for the M439 shall be 950 milliseconds to 20.275 seconds.

3.4.3.1.2 Launch cycle time. The time required to launch 2.75 inch rockets after receipt of command from the DU is as follows:

3.4.3.1.2.1 Non-fuzed rockets. The launch cycle time for firing non-fuzed rockets shall not exceed 40 milliseconds.

3.4.3.1.2.2 M433 fuzed rockets. The launch cycle time for firing rockets having M433 fuzed warheads shall not exceed 54 milliseconds.

3.4.3.1.2.3 M439 fuzed rockets. The launch cycle time for firing rockets having M439 fuzed warheads shall not exceed 64 milliseconds.

3.4.3.1.3 Accuracy. The OU fuze setter shall set RC time fuzes to the following accuracies:

M433 Fuze
PEN-M-Mode
Jungle Height (25-40 meters) ± 6% or 6 milliseconds whichever is greater.

Bunker Penetration (BNK) ± 6 milliseconds

M439 Fuze
RNG-KM ± 1.25% or 125 milliseconds whichever is greater.

3.4.3.1.4 Failsafe. Failure of any one component in the OU shall not cause firing of any rocket.

3.4.3.1.5 Fault detection. The OU shall have a Built-In-Test (BIT) capability to check the operational status of the unit. BIT shall obviate the need for any external test equipment for the OU at the aviation unit maintenance level (AVUM). The BIT shall functionally test the OU except for the firing circuits.

3.4.3.1.5.1 Self test. The OU will complete self test on command from the DU. Upon successful completion of the test, the Equipment Status Indicator (ESI) will be in normal or reset mode, displaying black on the indicator.

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3.4.3.1.5.2 Malfunction indication. An Equipment Status Indicator (ESI) shall be provided on the OU. The ESI shall be a go/no-go device that shall hold the last indication if power is removed. The ESI shall be visible on the outer surface of the unit when installed on the aircraft. The ESI shall also provide an electrical output for driving a remote indicator. This output shall be a switch closure to 28V standby power for a no-go condition and an open circuit for a go condition.

3.4.3.1.6 Proof test. Operations Unit Test Sets shall operate when assembled to an accepted AH-IS helicopter.

3.5 Environmental conditions. The Operations Unit shall operate and meet the performance requirements of 3.4.3 under any of the environmental conditions, or reasonable combination of these conditions, as specified below:

3.5.1 Natural environments. The operations unit shall be capable of operation and storage in the worldwide environments as specified below:

3.5.1.1 Altitude. Operating: 0-20,000 feet; nonoperating and storage: 0-50,000 feet.

3.5.1.2 Temperature.

Operations Unit.

a. Operating: -65° to $+160^{\circ}F$
b. Nonoperating and storage: -65° to $+160^{\circ}F$

3.5.1.3 Humidity. Operation after exposure to $149^{\circ}F$ and relative humidity to 95 percent.

3.5.1.4 Sand and dust. Operation after internal and external exposure to sand and dust test.

3.5.1.5 Fungus. Operating, nonoperating and storage after being subjected to fungus specified below:

Fungus	Serial Number
Aspergillus niger	9642
Aspergillus flavus	9643
Aspergullus versicolor	11730
Penicillium fumculosum	11797
Chaetomum globosum	6205

3.5.1.6 Explosive atmosphere. Operation in explosive vapor within or surrounding the Operations Unit.

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3.5.1.7 Rain. Operation without degradation in specified performance and shall sustain no physical damage during periods of heavy percipitation.

3.5.1.8 Service environment. The Operations Unit shall be designed and constructed to withstand the service environment as specified in MIL-E-5400 for Class 1A equipment.

3.5.1.8.1 Vibration. The Operations Unit shall not sustain damage or subsequently fail to provide specified performance when subjected to the vibration environment defined by MIL-E-5400, Figure 2, (Sheet 1), Curve III.

3.5.1.8.2 Shock. The Operations Unit shall meet the shock requirements of paragraph 3.2.24.6.1 of MIL-E-5400. The requirements of paragraph 3.2.24.6.2 of MIL-E-5400 for mounting hardware crash safety shall only apply to the units mounted inside the cockpit.

3.5.1.8.2.1 Bench handling. The Operations Unit shall withstand shock environments encountered in maintenance and service.

3.6 Workmanship.

3.6.1 Fabrication. Finished items and parts shall be fabricated in a thorough, workmanlike manner, and all manufacturing, processing, and assembly operations shall be correctly performed. Unauthorized salvaging operations (e.g., hammering to shape, repair by welding, straightening, bending, etc.) shall not be permitted.

3.6.2 Parts. The parts shall not exhibit faulty material and processing such as seams, laps, laminations, cracks, visible steps, sharp edges, nicks, scratches, burrs, deformations, unblended radii, chips, dents, rust, dirt, grease and oil (except where specifically required), corrosion products, flux, other foreign matter, and missing operations which may affect serviceability, operation, or safety.

3.6.3 Fasteners. Threads shall be full and undamaged for the entire minimum length or depth as required on the applicable drawing.

3.6.4 Cable assemblies and wires. Cable assemblies and wires shall be neatly dressed and tied in such a manner as to prevent pinching or chafing the wire insulation. Ends of wires shall be free from burrs, sharp edges or points.

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3.6.5 Castings. Casting shall be clean, sound, and free from blowholes, porosity and inclusion.

3.6.6 Welding. Welds shall be reasonably smooth and free of craters and cracks. They shall exhibit characteristics of good fusion penetration and soundness of weld deposit representative of good welding practice. They shall be free of irregularities which indicate lack of skill or experience on the part of the operator. All scale and splatter shall be completely removed prior to application of protective coatings. There shall be no cracks in or near welded areas.

3.6.7 Soldering. All soldering shall be performed in a thorough, workmanlike manner with no evidence of burning or charring.

3.6.8 Coatings. Exterior surface coatings shall be continuous; however, a few light scratches not exposing base material may be permitted.

3.6.9 Cleaning. The cleaning agents shall not be injurious to any part or assembly.

3.6.10 Packing. All required packing shall be dry.

3.6.11 Markings. All required markings and stamping shall be neat and sharply defined.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection and standard quality assurance provisions. Unless otherwise specified herein or in the contract, the provisions of MIL-W-63150 shall apply and are hereby made a part of this detail specification. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Certification of provisions. The certification provisions of MIL-W-63150 shall apply.

4.3 Classification of inspection. The following types of inspection shall be conducted on this item.

a. First Article Inspection.

b. Quality Conformance Inspection.

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4.4 First article inspection.

4.4.1 Submission. The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with provisions of 4.4.2. The first article sample shall consist of five Operation Units.

4.4.2 Inspections to be performed. First article assemblies components and test specimens may be subjected by the Government, at a place designated by the Government, to any or all of the examinations and tests specified in the item detail specification and to any or all of the requirements of the applicable drawings. See Table III specified herein.

4.4.3 Rejection. If any assembly, component, or test specimen fails to comply with any of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate its inspection upon any failure of an assembly, component, or test specimen in the sample to comply with any of the stated requirements.

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TABLE III. First article inspection

CLASSIFICATION OF DEFECTS & TESTS MIL-O-70483A(AR)

PARAGRAPH	TITLE: Operations Unit		SHEET 1 OF 1		DRAWING NUMBER: See Below
					NEXT HIGHER ASSEMBLY: ----
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE *UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Operations Unit (Dwg 9324108)				
	Examination for defects	5		3.4	4.5.2.1
	Interchangeability	5		3.4.1	4.5.3
	Proof test	5		3.4.3. 1.6	4.6.4
	Altitude	5		3.5.1.1	4.6.5.1
	Temperature	5		3.5.1.2	4.6.5.1
	Humidity	5		3.5.1.3	4.6.5.2
	Sand and dust	5		3.5.1.4	4.6.5.3
	Fungus	5		3.5.1.5	4.6.5.4
	Explosive atmosphere	5		3.5.1.6	4.6.5.5
	Rain	5		3.5.1.7	4.6.5.6
	Service environment	5		3.6.1.8	4.6.5.7
	Vibration	5		3.5.1. 8.1	4.6.5.7.1
	Shock	5		3.5.1 8.2	4.6.5.7.2
	Bench handling	5		3.5.1. 8.2.1	4.6.5.7.2.1

NOTES:

* The same five units shall undergo all the Examination or Test in the sequence listed.

DRSMC-OA (D) Form 160, 1 Aug 83 replaces edition of 1 Jul 77 which may be used until exhausted.

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4.5 Quality conformance (acceptance) inspection. Acceptance inspection shall be performed and shall be in accordance with the following:

4.5.1 Lot formation. The term "inspection lot" is defined as a homogeneous collection of units of product from which a representative sample is drawn or which is inspected 100 percent to determine conformance with applicable requirements. Units of product selected for inspection shall represent only the inspection lot from which drawn and presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer in one unchanged process, using the same materials and methods, in accordance with the same drawings, same drawing revisions, same specifications and same specification revisions and complies with the provisions for submission of product as specified in MIL-STD-105. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein, regardless of the type of inspection procedure which is being applied to determine conformance with the requirements.

4.5.2 Examination. Examination for the inspection listed in the following Classification of Defects and Tests paragraphs shall be performed by sampling to the AQL specified unless 100 percent inspection is indicated. Unless otherwise specified sampling shall be in accordance with MIL-STD-105 Level II, except the contractor may request permission from the procuring activity to use an equivalent continuous sampling plan from MIL-STD-1235.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.5.2.1	Operations Unit	1 OF 1	9324108
			NEXT HIGHER ASSEMBLY 9324106

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101.	Operations unit damaged (puncture, dented or deformed)		100%	3.4	Visual
102.	Assembly incorrect		100%	3.4	Visual
103.	Assembly incomplete		100%	3.4	Visual
104.	Component missing, damaged or improperly assembled		100%	3.4	Visual
105.	Operation missing or improperly performed		100%	3.4	Visual
106.	Nameplate label missing		100%	3.4	Visual
107.	Materials not as specified		100%	3.4	Visual
108.	Preconditioning/performance		100%	3.4.2/ 3.4.3	4.6.2
Minor					
201.	Nameplate marking misleading, illegible, incomplete or missing		1.5%	3.4	Visual
202.	Poor workmanship		2.5%	3.6	4.6.6

NOTES:

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4.5.3 Testing. The tests (First Article and Quality Conformance) listed herein, shall be performed on inspection lots as defined in 4.5.1 of this specification. The sampling plans shall conform to the provisions of 4.5.2 above.

4.5.4 Inspection equipment. The inspection equipment required to perform the inspections and tests prescribed in this specification is identified in the "Paragraph Reference/Inspection column in the tables, starting with paragraphs 4.5.2.1 and test methods paragraph (see 4.4). The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See 6.3 herein.

4.6 Methods of inspection.

4.6.1 Materials, components and processess. Compliance with all requirements of Section 3 of this specification shall be ascertained by current and continuing examination of inspection and test data to determine that all components (parts, sub-assemblies, and materials) have been inspected and tested and found to comply with their respective drawing and specification requirements, and that all specified manufacturing processess have been followed.

4.6.1.1 Visual and mechanical inspection. The visual, manual and mechanical inspections and tests shall verify compliance with the requirements of Sections 3 and 5 of this specification in accordance with 4.5.2 and Table III.

4.6.2 Performance test. The Operations Unit shall be tested for compliance with specified performance requirements after being subjected to the preconditioning requirement (see 3.4.2). Any M138 Operations Unit which fails to comply with the specified requirement or evidence of degradation of its operation shall be classed defective. Test equipment and procedure shall be satisfactory to the contracting officer (see 4.5.4).

4.6.3 Interchangeability (first article inspection only). The Operations Unit shall be subjected to the interchangeability tests specified (see 3.5.3). At the discretion of the Government representative, an interchangeability retest may be allowed. Failure during the retest shall cause rejection of the first article. A sample of five Operation Units from each retest shall be tested using the same procedures specified.

4.6.4 Proof test (first article inspection only). When specified in the contract (see 6.2) five (5) M138 Operation Unit Sets shall be forwarded to the site specified by the contracting agency and utilized on an AH-IS helicopter to determine that the test set will function properly within the system. The test sets shall function in accordance to the intent of the design and with no evidence of malfunction.

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4.6.5 Environmental tests (first article inspection only).

4.6.5.1 Temperature - altitude. The Operations Unit shall be subjected to the test of Method 504.1 of MIL-STD-810C for category 3 equipment with the following exceptions;

a. Use 20,000 feet where 15,000 feet is specified.

b. Do not exceed the temperature limits of 3.5.1.2.

4.6.5.1.1 High temperature. The high temperature test requirements are satisfied by 4.6.5.1. No separate test is required.

4.6.5.1.2 Low temperature. The Operations Unit shall be subjected to the test of Method 502.1, Procedure I of MIL-STD-810, except maximum low temperature to be -65°F.

4.6.5.2 Humidity. The Operations Unit shall be subjected to the test of Method 509.1 of MIL-STD-810.

4.6.5.3 Sand and dust. The Operations Unit shall be subjected to the test Method 510.1 of MIL-STD-810, Procedure I, with the exception that for equipment mounted enternal to the aircraft, the air velocity in Steps 1, 2 and 3 shall be 200 + 100 fpm. Step 3 may proceed immediately after stabilization in Step 2, if desired.

4.6.5.4 Fungus. The Operations Unit shall be subjected to the test of Method 508.1 of MIL-STD-810, Procedure I. Documentary evidance acceptable to the procuring activity that the material will not be nutrients for fungi may be used in place of the tests.

4.6.5.5 Explosive atmosphere. The Operations Unit shall be subjected to the test of Method 511.1 of MIL-STD-810, Procedure I, except test to be limited to maximum altitude of 20,000 feet.

4.6.5.6 Rain. Method 506.1, Procedure I of MIL-STD-810 shall apply. During each 30-minute rain period, the wind shall be maintained for 15 minutes commencing 5 minutes after the start of the period as specified in the procedures. The Operations Unit shall be operated during the last 10 minutes of the 30-minute period.

4.6.5.7 Service environment. The Operations Unit shall be tested in compliance with the requirement of section 3 of this specification in accordance with 3.5.1.8.

4.6.5.7.1 Vibration. The Operations Unit shall be subjected to Method 514.2-3 of MIL-STD-810, Procedure I, Category C.

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4.6.5.7.2 Shock. The Operations Unit shall be subjected to the Test Methods 516.2 of MIL-STD-810, Procedure I.

4.6.5.7.2.1 Bench handling. The Operations Unit shall be subjected to the Test Method 516.2 of MIL-STD-810, Procedure I.

4.6.6 Workmanship. The requirement of 3.5 shall be verified by continuous inspection of all parts and assemblies to assure conformance to high quality workmanship.

5. PACKAGING

5.1 Preservation, packaging, packing and marking. The operation unit shall be preserved, packaged, packed, and marked in accordance with the requirements of the Preservation and Packaging Data Sheet for P9324108 at the level of protection specified in the contract (see 6.2b).

5.1.1 Spare parts or replacement parts. Spare parts or replacement parts shall be cleaned, preserved, and packaged as specified in the contract or order.

5.2 Commercial packaging shall be in accordance with ASTM-D-3951.

6. NOTES

6.1 Intended use. The Operations Unit in quantities of two or four are combined with a Display Unit to form the M138 Rocket Management Subsystem which is intended for use on the AH-IS Cobra Helicopter to control the fuze setting and firing of 2.75 Inch Aerial Rockets.

6.2 Ordering data. Procurement documents shall specify the following:

6.2.1 Procurement requirements

a. Title, number and date of this specification.

b. Quantity required and delivery schedules.

c. Serialization requirements, if applicable.

d. Quality Conformance Inspection, if other than specified in Section 4 of this specification.

e. First Article sample requirements, if other than specified in Section 4 of this specification.

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f. Packaging requirements, if other than specified in Section 5 of this specification.

g. Certificates of conformance for each lot or shipment of product.

h. Location to which items are to be shipped.

6.2.2 Contract data requirements. Contract data requirements for inspection equipment design shall conform to Data Item Description DI-R-1714, Tailored.

6.3 Submission for inspection equipment design approval. See MIL-W-63150. Equipment designs shall be submitted as required to: Commander, AMCCOM, ATTN: AMSMC-QAF-I(D), Dover, NJ 07801-5001. Request letter of submittal shall state contractor, contract number, specification number, item nomenclature, and classification of defects and test paragraphs.

6.4 Submission of results of contractor-conducted examinations and tests. Unless otherwise specified by the Contracting Officer, the contractor shall forward requested records of examinations or tests to: Commander, AMCCOM, ATTN: AMSMC-QAF-S(D), Dover, NJ 07801-5001.

6.5 Drawings. Drawings listed in Section 2 of this specification, under the heading US Army Armament Research and Development Center (ARDC), may also include drawings prepared by and identified as US Army Armament Research and Development Command (ARRADCOM), Rock Island Arsenal or Picatinny Arsenal. Technical data originally prepared by these activities is now under the cognizance of ARDC.

6.6 Previous Issues. MIL-O-70483A(AR) is being released as a needed administrative correction to the previous issue and, therefore, contains no technical corrections or differences from the original version.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-O-70483A	OPERATIONS UNIT, FOR ROCKET MANAGEMENT SUBSYSTEM, M138

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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